

Speech Recognition Spring 2025 Project 7

Due date: March 04th 2026 3pm Beijing time

In this assignment, you will practice fine-tuning the Whisper model on an English speech dataset. You are required to complete the code at the specified locations and understand the core workflow of training Whisper.

You need to complete the following three stages in the provided code:

1. **Prepare the dataset for Whisper fine-tuning.**

We have provided the wTIMIT dataset at: `/dkucc/group/courses/compsci304-2526-s3/cl688/wTIMIT`, which contains both normal speech and whisper speech. The original split is not suitable for speech recognition training, as wTIMIT utterances are duplicated in the train and test sets. You should split the dataset into train/dev/test sets with a ratio of 400/25/25, based on the 450 sentences listed in `wTIMIT/doc/sentences.txt`. Train and evaluate only on whisper speech with a Singapore accent. For example, *SG/015/s015u313w* is a whisper speech sample with a Singapore accent.

2. **Post-train the Whisper base model on your prepared dataset.** Choose one training configuration from project5.

3. **Decode and evaluate the Word Error Rate (WER) on the test set.** Choose one decoding configuration from project6. Report the WER before and after fine-tuning to compare the effect of fine-tuning.

You should explain how you filled in the blanks in your report.